

packaging technology (which is shifting to glass substrates), and silicon photonics for continued progression of Moore's Law as the company's guiding light. Pressing the memory for more bandwidth, tightening the thermal envelopes, and packing in the silicon harder than ever before.

One of the ways Intel's going to grow and consolidate its position in the chip industry is by having "two bites at the apple," in Gelsinger's own words – hinting how Intel will have traditional chip products and newer foundry offerings to entice big tech players. "Am I ever going to get Google to say let's stop doing the TPUs (cloud tensor processing units) and use Intel's Gaudi line instead? Maybe, maybe not. Will I get them to be a foundry of ours? Okay. I have two shots to have many of those really big footprints come to us," according to him.

He also thinks that computing in general is moving from cloud native development to edge AI native development. "The fact that I can do big training models in the cloud, that's interesting. The fact that you can spend a billion dollars creating a multi-trillion parameter model, that's super interesting. But how do I enable hundreds of millions if not billions of people to use it? That's not going to happen by accessing one big cloud. That's going to be by pushing it out to the client and to the edge," Gelsinger said, believing that Intel has a great opportunity to be a true technology and product leader once again by leveraging these key trends.

Driving siliconomy in an uncertain world

Of course, being an American multinational corporation trying to claw its way back to the summit of chip manufacturing excellence isn't without its geopolitical challenges. With the world still recovering from global chip shortage and supply chain constraints of the 2020 pandemic, and knives being sharpened over impending chip wars, Intel is preparing to adjust to an increasingly uncertain world as best it can. "We want to build, I'll say, an American supply chain,

a European supply chain, and an Asian supply chain that is reasonably unique and valid for those markets," emphasises Gelsinger as key pillars of Intel's strategy, where there's independence and interdependence between these supply chains. "We're doing in Germany and Poland, combining what we already do in Ireland and Israel. We're also emphasising in Eastern Europe through more of the systems and components assembly, particularly in the Czech Republic and Romania. This will bring autonomy and independence for the European supply chain, so they're more resilient," he said.

Is that going to satisfy all the needs of Europe? Absolutely not. But will it make them more resilient for disruptions to occur? Absolutely yes, according to Gelsinger. Similarly, the American supply chain is going beyond Oregon to places like Arizona, New Mexico, and Ohio for packaging related supply chains. "Will it satisfy all of their market needs? Absolutely not. Will it make them more resilient and capable of surviving disruptions? Absolutely yes. And similarly, you know, the Asian supply chains are pretty well established, but even there we're investing more in Malaysia. We are working with key ODMs, component suppliers, mineral suppliers, chemical suppliers, and the like," he highlighted.

Intel is doing all of this to ensure they don't have any single point of enormous dominance or risk associated with any one port, country, or supplier. "Because if this truly becomes a trillion dollar plus industry that fuels the entire global economy, there should be no single point of failure, no single point of dependency," according to CEO. There'll still be economic disruptions that occur, either from a natural calamity to a geopolitical conflict somewhere, but Gelsinger believes that the world must be fundamentally more resilient for something so important as their technology supply chains going forward.

ARM and NVIDIA enemies no more

A crucial piece of realignment Intel has done for its foundry aspirations to flourish

is to no longer view traditional competitors with a single lens. If Intel has to produce chips for the likes of Qualcomm or NVIDIA in the future, then it needs to find innovative ways to partner with them, something that Gelsinger has emphasised since returning to Intel in 2021.

According to Gelsinger, Intel's future is full of leadership in driving open standards, leadership in products and leadership in technology – out of which has emerged the ARM partnership. Championing the cause of RISC, the ARM architecture has given Intel's x86-based CISC platform a huge run for its money, ever since the proliferation of smartphones from the late 2000s – all of which are ARM-based. Needless to say, Intel needs ARM to grow its foundry business.

"The majority of the wafer usage in foundries today and well over 50 percent of the world's foundry revenue is based on ARM," said Gelsinger, in no uncertain terms. "We viewed ARM as a competitor in the past, but now they're a critical collaborator since we announced a partnership back in April 2023 and I'll just say it's going very well. We were one of the foundational investors in their IPO recently, and we're getting very good results from the preliminary design work we're doing on Intel 18A process node," he continued.

Gelsinger also mentioned NVIDIA as another traditional rival – in the AI space – that's potentially an Intel foundry customer of the near future. Acknowledging NVIDIA as the incumbent to challenge in the rapidly accelerating AI platforms market, Gelsinger said no other company was better prepared to capture the AI opportunity than NVIDIA, largely down to Huang's genius – whom he has known for over three decades. Of course, it wasn't without a bit of luck. **d**

